Tim Kane

Key points

- Foot-and-mouth disease is not present in Australia.
- It has recently been detected in neighbouring Indonesia for the first time in 30 years.
- Border settings have been adjusted to keep Australia foot-and-mouth disease free as Indonesia gets this under control.
- While unlikely, an incursion would cause substantial economic loss to the agriculture sector and social harm in affected communities.
- The most substantial economic cost of an incursion would be loss of access to export markets.
- Foot-and-mouth disease is not a food safety issue.

Australia's world-class biosecurity system has kept the country foot-and-mouth disease (FMD) free for 150 years. It is expected that Australia's FMD-free status will continue. However, Australia has been on higher alert after an outbreak of FMD was reported in cattle in Indonesia in May 2022. It was subsequently detected in the Indonesian province of Bali in July.

FMD is a highly contagious disease which affects cloven hoofed animals including cattle, sheep, goats, pigs, deer, buffalo and camelids. It can spread through domestic and feral animal populations. While it spreads through animal populations through close contact with infected animals, it can also be spread through contaminated clothing, footwear or equipment. However, it is not a food safety issue and does not pose a risk to humans.

Prior to the outbreak in Indonesia, an estimated 77% of global livestock were in regions where FMD is known to circulate. Australia adopts a responsive and risked based approach to biosecurity, with risks managed pre-border, at the border and post-border (within Australia). In response to the detection of FMD in Indonesia, Australia's already robust biosecurity measures were further calibrated to prevent FMD from entering the country. This includes the introduction of sanitation mats in airports, additional biosecurity officers, more targeted inspections and stronger clearance requirements for travellers entering Australia. Australia is also working closely with Indonesia to assist in managing its outbreak, including through providing funding for vaccinations.

Australia is free of FMD, and since the outbreak, Indonesia has been making good progress on getting it under control. However, if an outbreak were to occur in Australia, the economic consequences for livestock production and exports could be severe. FMD is a World Organisation for Animal Health (WOAH)-listed disease with clearly articulated international guidelines and standards for diagnosing, controlling and managing an outbreak to regain FMD free status, in all or part of the country. FMD represents a substantial threat to Australia’s livestock industries and export markets. An outbreak would jeopardise the export of all cattle and other cloven-hoofed animals and their products. FMD has the potential for rapid and extensive spread, but early detection and response could limit the spread and duration of any outbreak.

In March 2022, Lumpy Skin Disease (LSD) was reported in Indonesia. It is also present in neighbouring countries, such as Malaysia, Singapore, Vietnam and Thailand. It was first detected in the region in 2019 and continued to spread. LSD affects cattle and water buffalo.

Australia is free from LSD. Like FMD, if LSD were to make an incursion into Australia it has the potential to cause substantial harm to Australia's cattle (meat, meat by-products and dairy) industry. Given its recent incursion into Indonesia, the proximity of LSD is a concern for Australia. As with FMD, Australia is providing support and funding to help control the outbreak of LSD in Indonesia.

The transmission of LSD is primarily through biting arthropods, such as mosquitos, flies and ticks. It may also be possible for LSD to enter via an insect carried to Australia by wind from a nearby country. This means there are different challenges to preventing, containing and eradicating an outbreak of LSD (when compared with an FMD outbreak).

Depending on the size and spread of an outbreak, in addition to animal health, welfare and productivity impacts, the community impacts and the economic costs of LSD could be severe. As with FMD, the most significant costs would be the result of losing access to overseas markets. The attitudes of trading partners toward an outbreak of either FMD or LSD will be contingent on a number of factors, including how widely it spreads and the measures that Australia would employ to contain and eradicate the diseases.

Given the heightened attention to FMD, it is important not to lose sight of other biosecurity threats which are still a concern, such as LSD. While the rest of this paper considers the impacts of FMD, it does not seek to dismiss the serious concern also raised by the risk of incursion of LSD, or a range of other persistent threats currently being actively managed by Australia's biosecurity system.

In the event of an outbreak of FMD, governments, agencies and industry would seek to contain, control and eradicate the disease, using well-rehearsed response plans, to re-establish the FMD-free status of Australia as quickly as possible, while minimising social and financial disruption. The AUSVETPLAN for FMD provides guidance on a nationally agreed approach to support decision making in the event of a potential FMD incursion. The plan considers a wide range of possibilities in the event of an incursion, given the different scenarios under which an outbreak could occur.

If a case of FMD is diagnosed in Australia, or there is a strong suspicion of the disease, a national livestock standstill would be rapidly considered with all jurisdictions to contain the outbreak by preventing the movement of all FMD-susceptible animals. These movement restrictions could also prevent the movement of animal products and equipment and would be imposed for at least 72 hours. During the livestock standstill, assessments would be made of the relative risks of FMD spreading across different regions of Australia. This would inform the next steps of the response to minimise the impact, stop the spread, and ultimately shape the eradication strategy employed.

When the national livestock standstill is lifted, the restrictions placed on farms and processors would depend on the risk classification assigned to each premise and the area. The disposal of livestock and livestock products from high-risk farms may occur. However, in low-risk areas, movement of livestock products from farms to approved processing facilities could generally be allowed, so long as certain criteria outlined in the agreed emergency response plan are met. Jurisdictions may also impose their additional requirements, depending on their individual circumstances.

Strategies to contain and eradicate FMD and begin on a trajectory to demonstrate FMD-free status, could include quarantine and movement controls, stamping out of infected and suspect animals, disease surveillance and strategic use of vaccination.

Livestock zoning (or regionalisation), which is where FMD-free zones are established under strict conditions, may be considered as a strategy for restoring access to international markets in some cases. In the event of a contained detection of FMD in Australia, the Department of Agriculture, Fisheries and Forestry (department) would commence negotiations with trading partners on acceptance of regionalisation. The nature and location of an incursion would inform the department’s submission to WOAH and the approach to trading partners. However, the submission would need to be provided with substantiated evidence gathered over a defined time period to support consideration of zoning. It should also be clearly noted that the department anticipates that trading partners are likely to request reciprocal recognition of their proposed arrangements, requiring additional departmental resources to assess those requests.

In relation to vaccinating animals, vaccination prior to an outbreak would risk the FMD-free status of a country, and trigger many of the same restrictions on trade which Australia would seek to avoid from an outbreak. If vaccination is part of the eradication strategy, any vaccinated animals would need to be destroyed prior to full FMD-free status being granted by WOAH.

WOAH is the World Trade Organization reference organisation for standards relating to animal health. WOAH sets international standards for diagnosis, control and prevention of animal disease including FMD. WOAH also provides guidance for safe international trade in livestock and livestock products. There are established procedures for regaining recognition of FMD-freedom by WOAH following an outbreak.

At a minimum, WOAH will require three months from the disposal of the last animal killed using a 'stamping out' policy without vaccination, and an application to demonstrate appropriate steps have been taken to eradicate, monitor and protect from further incursion.

However, following an outbreak and separate to the assessment and WOAH's official recognition of reinstatement of freedom from FMD, Australia would need to negotiate regaining market access with its trading partners. The timeframe for the recommencement of trade will be dependent on the nature of the outbreak, control program used, response of our animal health system, recognition of freedom by WOAH and the importing country’s own FMD freedom recognition guidelines. In order to recommence trade, the department will need to submit for the consideration of a trading partner, a comprehensive submission outlining the action undertaken, surveillance and other animal health data, traceability information (including current modernisation measures if completed) and national systems controls (among other references). This will require time to prepare and significant internal, state and territory representative and industry engagement.

Historical analysis of FMD outbreaks, suggest that even if an FMD outbreak is controlled quickly (such as in the case of Ireland which was within 2 days), the fastest anticipated FMD freedom recognition by a trading partner (the United States as an example) was 228 days. The longest time taken to recommence trade after an outbreak was in Japan, where trade to the United States did not recommence for 774 days.

As with the case for responding to any FMD incursion, the economic impacts could vary widely depending on the number of areas affected and the effectiveness of the response and eradication strategy. Prior modelling by ABARES explored different scenarios. In smaller outbreak scenarios which were modelled, Australia was quicker to eradicate the disease and regain market access. In the worst-case, large multi-state outbreak scenario, which was modelled, it would take longer to eradicate and substantially longer to regain market access than the smaller outbreaks and could cost the Australian economy $80 billion over ten years. In all scenarios, the cost would be overwhelmingly due to losing access to Australia's high value livestock export markets. Awareness of the disease and early detection of initial infected properties would limit an incursion to a much smaller outbreak, reduce time to eradicate and regain market access, and lead to a significantly smaller direct economic impact.

The WOAH Terrestrial Animal Health Code recommends that importing countries have controls in place to prevent movement from affected countries of potentially infected animals or animal products. In the event of an incursion, trading partners may undertake their own risk assessments for imported commodities and evaluations of Australia's FMD status to satisfy their own appropriate level of protection. It would mean that Australia's livestock exports would cease until appropriate evaluations are undertaken. Restoring trade takes time and will vary among Australia's trading partners.

To put the value of Australia's livestock export markets in context, Figure 1 shows the value of the top five export markets for Australian livestock and livestock products (including meat, wool, dairy and animal by-products) for the three years to 2021-22. This is not intended as a direct measure of potential economic cost, rather to outline that Australia's export markets are valuable and investment in their protection is vital.

Figure 1 Real value of livestock product exports, major trading partners, 2019-2020 to 2021-22

Even after market access is regained, there could be a risk that consumers' preferences in foreign markets, particularly those in high-value markets, may change as a result of the incursion and perceptions around the food safety of Australian products. Australian agricultural production currently maintains a reputation internationally as high-quality. It receives a premium in many markets based on this reputation. While FMD is not a food safety issue, FMD being associated with Australian animal and animal products exports may damage this reputation and reduce demand in international markets.

There are also direct costs associated with control measures, such as culling infected and high-risk animals, vaccinating animals, and decontaminating farms and equipment (for which there are measures designed to help compensate producers). In addition to the direct costs there would also be down-stream impacts across the supply chain, including for feedlots, transportation, processing and distribution. Downstream impacts would vary depending on the characteristics of regions affected and the size of an outbreak.

In the event of an outbreak exporters would suddenly be unable to sell livestock products in most, if not all international markets. There would be an immediate and large increase in the supply of livestock products to domestic markets. Australia exports around three-quarters of its livestock products which would mean the existing domestic market would be unable to absorb the increased supply, particularly in the short term. Even in the case of a small, localised outbreak, if all the beef currently destined for export markets was diverted to the domestic market, it would represent an increase of 270% in the volume currently supplied to the domestic market.

After initial shortages caused by restricted movement of livestock products domestically, the result could be a substantial fall in the domestic farmgate price of livestock and the retail price of meat. The domestic farmgate price of cattle and sheep would be expected to fall substantially over the first few months of an outbreak. Prices could fluctuate rapidly and change as participants in the supply chain adjusted their production decisions.

The short-term oversupply of meat in the domestic market and resulting price reductions would be moderated to the extent that producers hold back livestock from slaughter. Producers would likely respond differently to such a large price fall, with decisions to hold or sell their livestock depending on a number of factors. These factors include pasture and feed availability, meat processing capacity and access, the size and nature of herd and flocks, availability of transportation, the willingness of producers to sell their products at reduced prices and expectations of how quickly market access might be restored.

After the price shock in the first few months, and once producers had started to respond to the oversupply, the domestic price would be expected to increase from the low resulting from the shock. As turn-off of cattle and sheep decreases and production falls, the price of meat would begin increase depending on how much confidence producers in unaffected areas have of regaining at least some market access quickly. The greater the confidence, the more they will hold onto livestock. While the price would increase, it would remain below what it would have otherwise been with no incursion. ABARES modelling from 2013 modelled two small outbreak scenarios which estimated that 12 months after the incursion the price to be 12-16% lower than it would have been otherwise, under relatively favourable timelines for restored market access.

While pigs are also susceptible to FMD, there would be less of a direct impact to the supply of pig meat in the Australian market, as Australia exports less than 8% on average of what it produces. Similarly, only around 3% of chickens, which are not susceptible, are exported on average. The price impacts of these products are uncertain. Notwithstanding the potential for supply chain constraints caused by restricted movement of livestock to cause some localised shortages, the price of pig meat and chicken meat could fall, as consumers substitute to lower-priced beef and sheep meat. However, if consumers instead seek to buy more of other sources of protein, prices could instead rise.

Other livestock products would also be affected, however, there may be a different dynamic to how markets respond. For example, wool would be affected by export market access issues, as it is largely sold into international markets. Very little shorn wool is sold into the domestic market. Wool producers would have the option to consider storing wool until access to international markets are restored.

Source: https://www.agriculture.gov.au/abares/research-topics/agricultural-outlook/foot-and-mouth